

Why the Paper-Faithful Deep Snake Underperforms on FLAME-1

An investigation: what we tried, what each fix bought, and why the snake head fails

CS269 Project — Deformable Models

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1 Summary

The paper-faithful `deep_snake_paper` (CenterNet detector $\rightarrow$ octagon-from-box $\rightarrow$ circular-conv snake) scored far below the detector-free `deep_snake_simple` (0.658 IoU) and the other learned methods on FLAME-1. We ran a sequence of targeted fixes, each of which improved the *pipeline* but none of which made the *learned snake head* beat the simple octagon-from-box geometry it is built on. The final diagnosis: the offset-based snake cannot fit small, irregular fire contours — it fails to overfit even four fixed training samples — so the refinement step is a net negative on this data. The box $\rightarrow$ octagon prior does the real work.

2 The fixes we tried, in order

Table 1: Each intervention and its effect on FLAME-1 test IoU (301 frames). “Oracle” = snake given GT boxes; “System” = full detector $\rightarrow$ snake pipeline.

#	Intervention	Result	Verdict
0	Baseline (whole-frame box, conf 0.30)	System 0.050	broken
1	Lower detector conf threshold 0.30 $\rightarrow$ 0.10	System 0.050 $\rightarrow$ 0.218	helped (recall)
2	Per-instance boxes (one box/blob)	Oracle 0.096 $\rightarrow$ 0.497; System 0.218 $\rightarrow$ 0.348	helped a lot
3	Stride-4 features + scaled offsets	snake moves 1.3px $\rightarrow$ 6.4px; IoU flat	no net gain
4	Chamfer (anti-collapse) loss, weights 0.05–5.0	still collapses	failed

Fix 1 — detector confidence threshold (free, no retrain)

The detector emitted low-confidence boxes; the default 0.30 cutoff discarded most of them. Sweeping the threshold:

conf	0.30	0.20	0.15	0.10
fired %	24	60	83	96
mean IoU	0.050	0.125	0.181	0.218

IoU-given-a-detection stayed flat ($\approx$ 0.21) the whole way, so the recovered boxes were genuine detections, not noise — pure recall recovery, $4.4\times$ for free. *But this only fixed coverage, not per-detection quality.*

Fix 2 — per-instance boxes (the big one)

FLAME-1 fire is $\sim$ 9 scattered blobs/frame. The original code trained and evaluated the snake with *one bounding box over all the fire*, which is 98% empty (fire fills a median 2% of it). A single closed octagon cannot wrap nine disconnected specks. Switching to **one box per connected component** (a new `PaperSnakeDataset`, one training sample per blob) lifted oracle IoU from 0.096 to **0.497** ($5.2\times$) and system IoU to 0.348. This confirmed the dominant problem was the single-octagon-over-many-blobs mismatch, not the detector.

Fix 3 — the snake was a no-op

We then found the learned snake was barely moving: mean vertex displacement **1.27px** on a 512px image, and the raw octagon scored *higher* (IoU 0.488) than the snake’s final output (0.473). Two architectural bugs:

- **Stride-32 features.** The ResNet18 backbone (full, -2) gave a 16×16 feature map for a 512px image; an ~ 80 px box covered ~ 2 feature cells, so all 128 vertices sampled (nearly) the same feature and predicted identical near-zero offsets. We truncated the backbone at **layer1** (stride 4, 128×128 map).
- **Sub-pixel offsets.** Raw offsets were ~ 0.24 px; we added an offset scale ($\times 32$) so iterations can actually deform the octagon.

After retraining, vertices moved 6.4px — but IoU stayed flat and the contour now *shrank to $\sim 51\%$ of the octagon’s area*. It was moving, but moving inward (collapsing), not tracing the boundary.

Fix 4 — anti-collapse loss (failed)

We added a symmetric Chamfer term (**gt** $\rightarrow$ **pred** direction penalizes uncovered GT boundary points, i.e. collapse) to the cyclic-L1 contour loss. A unit test confirmed it heavily penalizes collapse (a shrunken contour costs $57\times$ a spread one). Yet across weights 0.05, 0.2, 0.5 (and later 1.0, 5.0) the snake *still collapsed* (final/octagon area 0.53–0.68; final IoU still below octagon-only).

3 Root cause: the snake cannot fit the target

We ran the decisive test — can the model **overfit four fixed samples**? A model that cannot overfit four examples has a capacity/optimization bug, not a data or regularization problem.

- Loss *does* decrease ($54\rightarrow 4.9$) and gradients flow (snake grad-norm ~ 2026), so wiring is fine.
- The target is good: rasterizing the GT contours themselves gives IoU **0.881** — a high ceiling.
- The loss correctly prefers tracing over collapse on a clean synthetic example (1.04 vs 122 at weight 0.05).
- **Yet on 4 fixed real samples, 200 steps, even at Chamfer weight 5.0, the contour collapses (area ratio 0.24–0.40).**

Mechanism. `get_octagon_from_box` produces a fixed-topology octagon with 128 evenly-spaced vertices. The snake predicts per-vertex offsets sampled from a feature grid. For a small, jagged fire blob there is no smooth offset field mapping evenly-spaced octagon vertices onto the irregular GT boundary — the vertex-correspondence problem is not solvable by local feature sampling. The optimizer’s best *reachable* move is to pull vertices toward the dense interior (lowering mean L1), which manifests as collapse. This is a known limitation of offset-based contour models on small/fragmented targets — precisely FLAME-1’s fire.

4 Conclusion

1. Every fix improved the *pipeline* (recall $0.05\rightarrow 0.22$; per-instance oracle $0.10\rightarrow 0.50$), but the *learned snake head never beat the octagon-from-box prior* (final ≈ 0.46 vs octagon ≈ 0.51).
2. On FLAME-1’s small, fragmented fire, the offset-snake refinement is a net negative: the box $\rightarrow$ octagon geometry is the better predictor.
3. This is an honest negative result for the paper-faithful design *on this data*. A non-offset refinement head (e.g. a per-instance mask head, or a boundary model with adaptive vertex count) would be the path forward; the offset-snake is the wrong tool for many tiny irregular regions.

4. Practical recommendation: use `deep_snake_simple` (0.658) or U-Net (0.772) for FLAME-1; if the paper pipeline is required, report octagon-from-detector-box (the snake’s input) rather than the collapsing snake output.

Artifacts: `results/rerun_flame1_20260601_193349/` (threshold sweep, viz), `results/rerun_flame1_perinstance_094117/`, `rerun_flame1_snakefix_1000`
Code: `flame/deep/dataset.py` (`PaperSnakeDataset`), `flame/deep/deep_snake.py` (stride-4 backbone, offset scale), `flame/deep/losses.py` (`snake_contour_loss`).